

Toward Real-time Generation with MiniMax H3

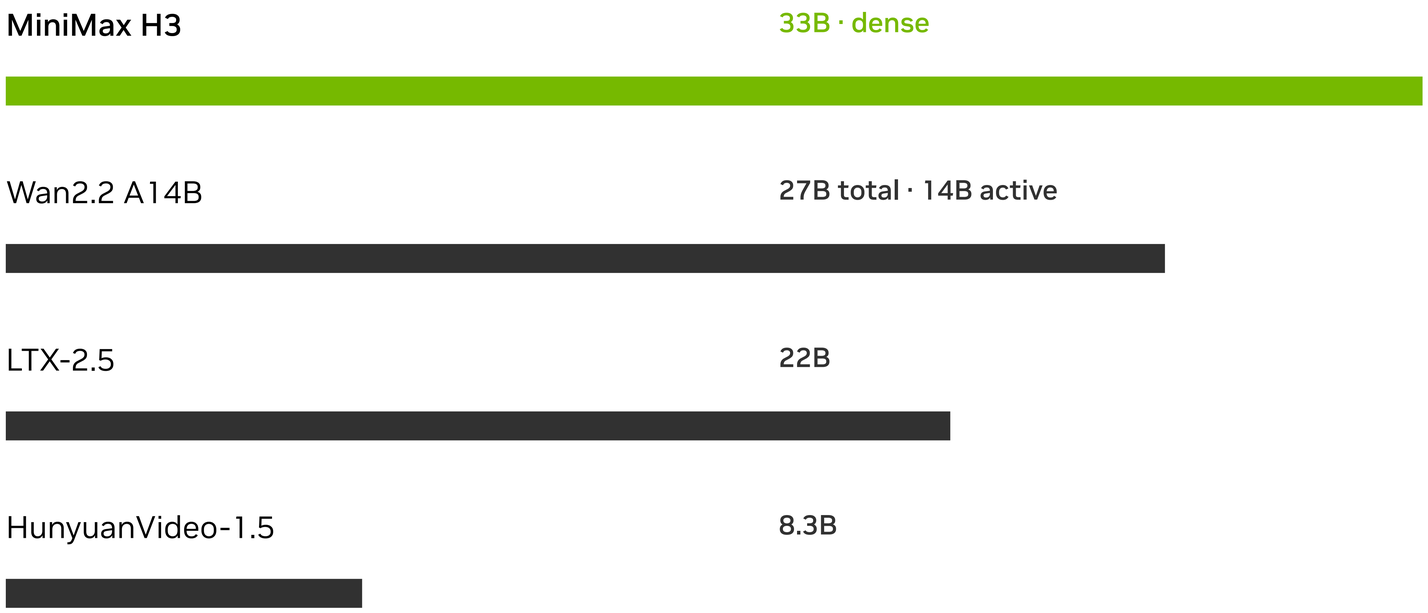
Li Shunyang Devtech Compute NVIDIA

2026-08-23

MiniMax H3 in Context

A large dense backbone for joint audio–video generation.

Generative backbone parameters



Audio + video

Generated together

Multimodal context

Text, image, video and audio

Native 768p

Open H3-Base capability

The challenge: fit the complete audio–video pipeline into a ~5 s budget.

Published full backbone sizes; encoders excluded. Parameter count is not a latency or quality ranking.

NVIDIA

SM120 Cards

Compute-rich, bandwidth-modest

RTX PRO 5000

×8

SMs	Memory	Power	Interconnect
110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16

Blackwell Tensor Cores

Native FP4 compute

No GPU NVLink

64 GB/s PCIe per GPU, one way*

*Raw nominal Gen5 ×16 capacity. The constraint here is inter-GPU communication relative to on-card compute.

NVIDIA

Baseline: Latency Breakdown

960 × 544 · 5.167 s generated video

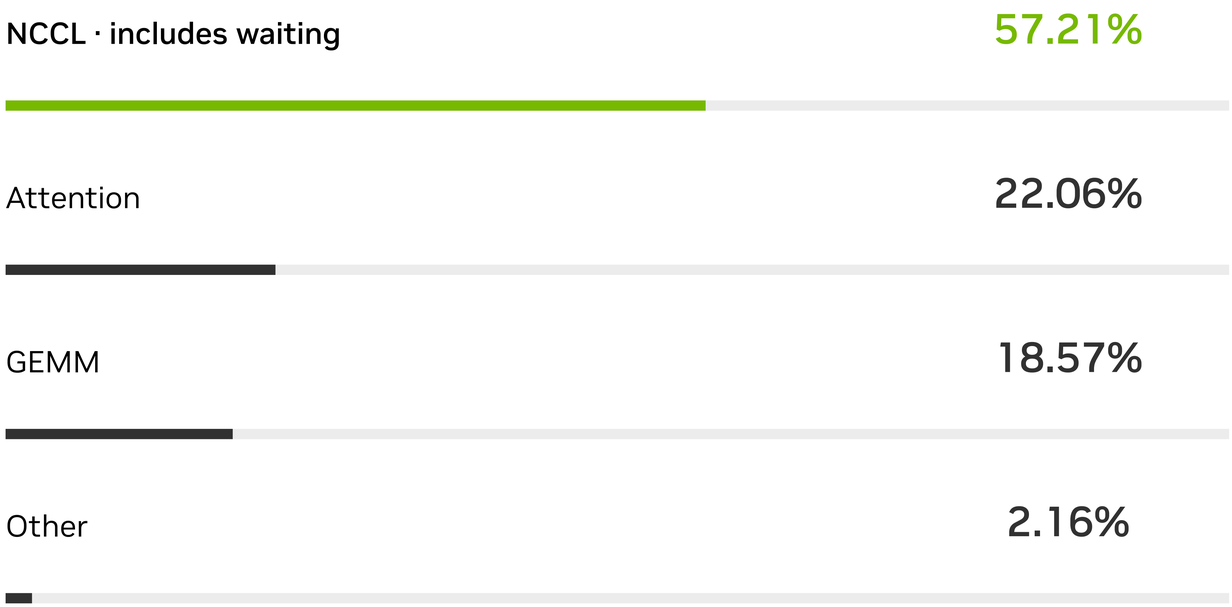
Reference configuration

213.4 s

E2E latency

- 8 × RTX PRO 5000
- 50 steps · BF16 · dense attention
- Ulysses8 · encoder TP8 · VAE tiles8

GPU work by operator



GPU work includes overlap and waiting; these shares are not E2E wall-time components.

NVIDIA

Optimization - Distillation

Reduce how often the model runs.

Base H3

50 model evaluations



84%

fewer evaluations

Distilled student

8 model evaluations



213.40 s → 44.33 s

Recorded E2E · 4.81×†

A trained student predicts multiple diffusion intervals per call.

Fewer evaluations reduce compute and communication together.

† Baseline measured separately. This is a recorded median ratio, not an isolated step-count experiment.

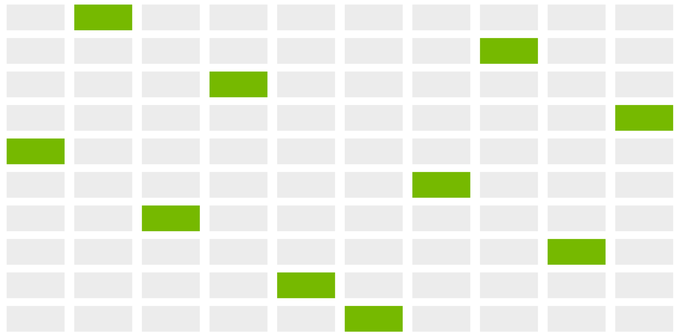
NVIDIA

Optimization - Video Sparse Attention + NVFP4

Do less attention work. Use lower precision where quality allows.

Video Sparse Attention

Coarse attention selects fine regions



Content-dependent selection · schematic

90% sparsity · Block size: 128 tokens

44.33 s → 25.92 s · 1.71×

NVFP4

QAD DiT + native FP4 attention

BF16 → NVFP4

Train for quantization error.
Keep sensitive paths at higher precision.

25.92 s → 19.46 s · 1.33×

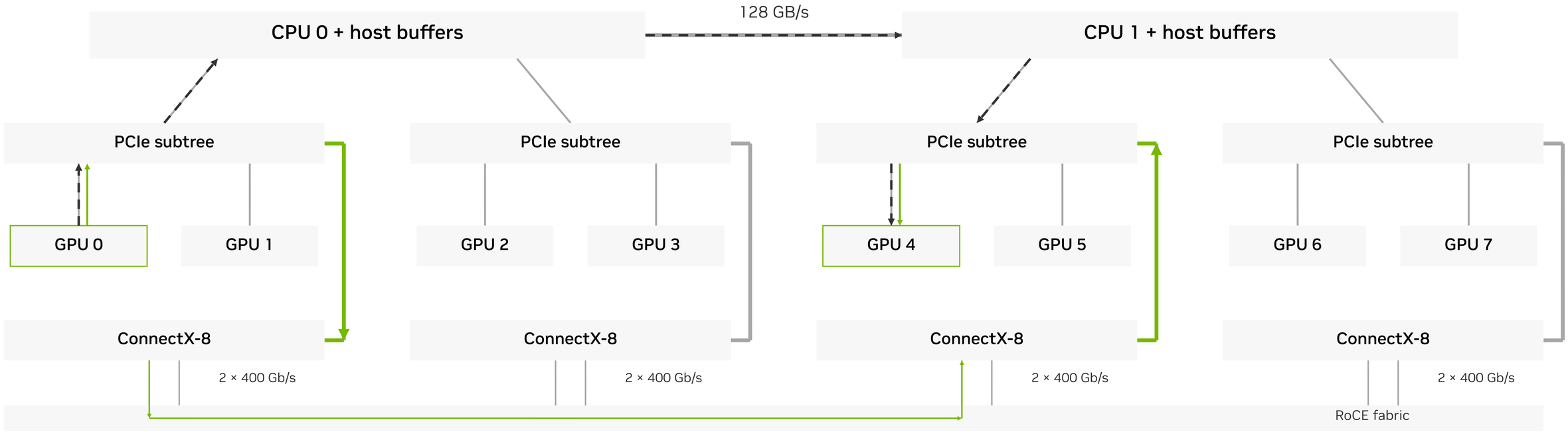
Evaluate speed and quality together on representative scenes.

Recorded E2E medians · 8 evaluations · trained sparse / quantized model variants; quality must be validated.

NVIDIA

Optimization - Comms

Compute-rich, bandwidth-modest: choose the payload route.



Nominal one-way links: GPU / host 64 GB/s · NIC PCIe 128 GB/s · port 50 GB/s

Original code · local-only NCCL replay

17.0 GB/s/GPU

5.560 ms · shared uplinks + host buffers

2.34×

GPU-affine FI RDMA

39.9 GB/s/GPU

2.375 ms · bypasses host payload buffers

Measured 8-GPU Q/K/V/O replay · S=15,104 · BF16 · 90 samples/backend · includes layout and synchronization.

Arrows illustrate GPU 0 → GPU 4. Rates are effective collective TX per GPU; historical Base50 pair routes were not recorded.

NVIDIA

Optimization - Comms

Choose the path, overlap the work, and reduce movement.

Route

GPU-affine NICs + BF16 RDMA.
Bypass shared host payload paths.

Overlap

2 / 3 / 2 head bands.
Transfer one band while another computes.

Streamline

Direct layouts + compute CUDA graphs.
Reduce copies and launch overhead.

Overlap transfers with attention



Conceptual overlap · 2 / 3 / 2 head bands

3.1× faster

19.46 → 6.83 → 6.41 → 6.19 s

Route/layout → head bands → compute graphs

Route efficiently. Overlap work. Reduce movement.

Grouped result: RDMA + layout + head overlap + compute graphs.

NVIDIA

From Minutes to Real Time

960 × 544 · 5.167 s output · 8 × RTX PRO 5000

Stage	Cumulative optimization	E2E latency	Speedup[1]
00	Baseline · BF16 / dense / 50 evaluations	213.402 s	—
01	+ PDD · 50 → 8 evaluations	44.327 s	4.814×†
02	+ Trained video sparse attention	25.915 s	1.710×
03	+ QAD NVFP4 DiT / native FP4 attention	19.460 s	1.332×
04	+ RDMA / direct layout / V-gate overlap	6.832 s	2.848×
05	+ 2 / 3 / 2 head pipeline	6.407 s	1.066×
06	+ Compute CUDA graphs	6.193 s	1.035×
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420 s	1.143×
08–12	+ Output fusion / routing / copy removal	4.922 s	1.101×

43.4× faster · 4.92 s · RTF 0.95

[1] Previous displayed row ÷ current row. 08–12 is grouped. † Baseline refreshed separately (n=2); later stages n=3.

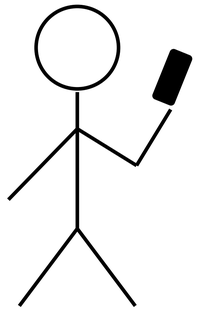
Standalone H3; ASR / LLM excluded. Includes model, schedule and precision changes. No new measurements.

NVIDIA

From a Question to a Reply

Voice → text → prompt → audio + video

"Hi, could you introduce yourself?"



User

Voice

Qwen3-ASR

0.6B · Speech → text

Qwen3-4B

Reply → H3 prompt

Reference image



H3

Audio + video

"Hi, I'm Miles..."



Avatar reply

PROMPT EXCERPT

... "Hi, I'm Miles. I help with simple, clear answers." ... His lips follow the words.

Hear the avatar's reply

960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

NVIDIA

Live Demo

A continuous conversation · final 75% at 3× playback



960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

NVIDIA

Takeaways

01

Optimize the communication path.

Compute-rich GPUs need topology-aware communication.

02

Validate quantization case by case.

Measure quality and latency together.

03

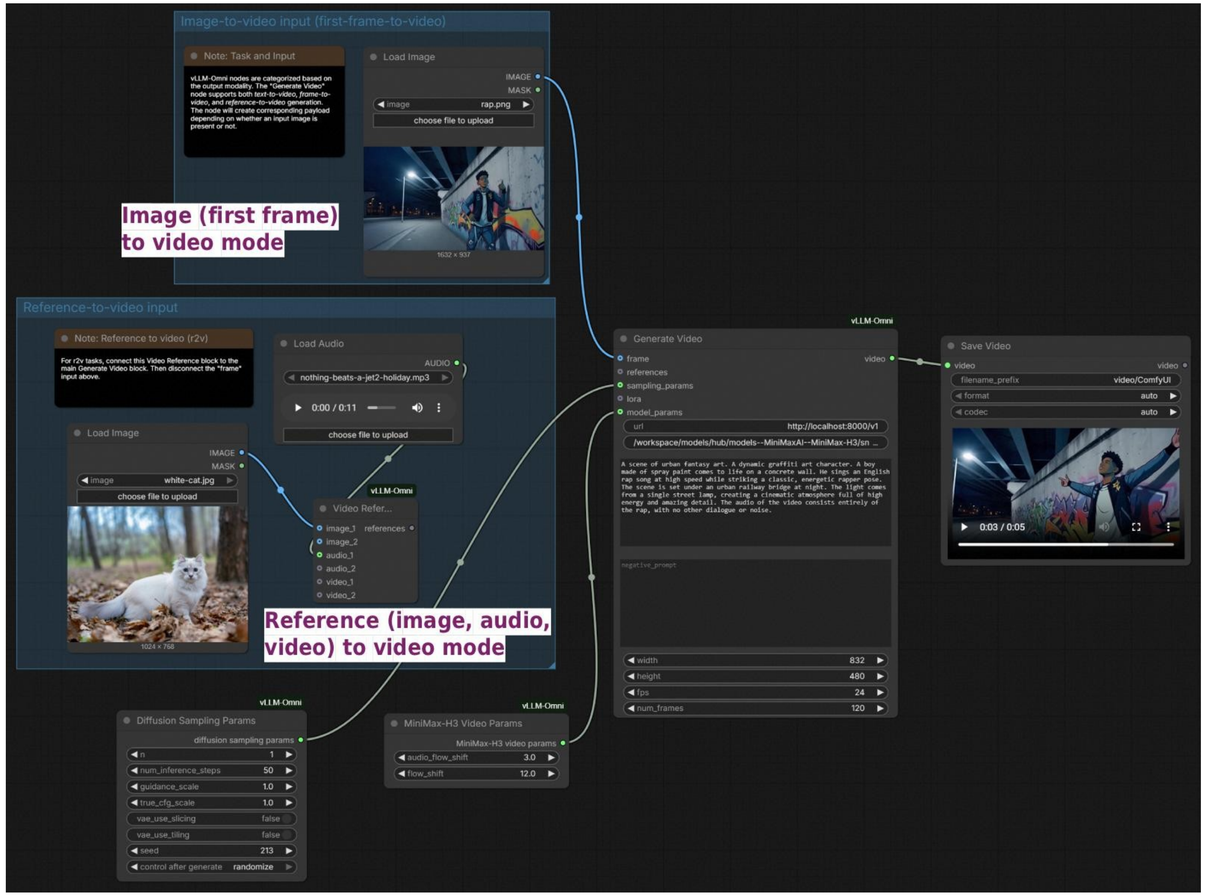
Next: responsive, richer interaction.

Autoregressive diffusion + prompt enhancement.

NVIDIA

vLLM-Omni for Creators — ComfyUI

Visual workflows, powered by a local or remote model service.



Compose in ComfyUI. Run models in vLLM-Omni.

AVAILABLE ON MAIN

Compose visually

Image, video and audio workflows.

Serve separately

Models run in vLLM-Omni.

Bring H3 into the graph

Text, keyframes or reference media.

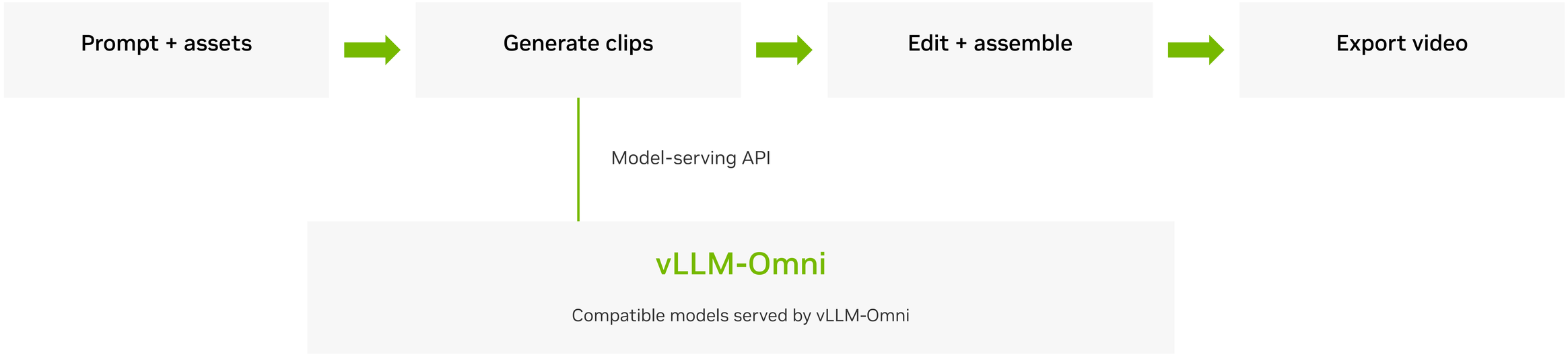
Official UI example, not benchmark settings. Upstream main verified 30 Oct 2026 · 08177216

NVIDIA

vLLM-Omni for Creators — Video Studio

Planned open-source release

A developer tool for video generation and editing.



One workflow, from generated clips to an edited video.

Video Studio is a working title. Diagram shows the planned workflow and backend integration.

vLLM-Omni Networking Hour!

https://github.com/vllm-project/vllm-omni

https://blog.vllm.ai/2025/11/30/vllm-omni.html

https://communityinviter.com/apps/vllm-dev/join-vllm-developers-slack

SCAN ME TO JOIN THE WECHAT GROUP

vLLM | vLLM-Omni